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Review

Privacy emotion recognition in artificial intelligence: Research frontiers, key challenges, and prospects

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ABSTRACT

With the arrival of the new wave of artificial intelligence, emotional intelligence has broad application prospects. By quantifying the psychological state of humans through psychophysiological calculations, machines can be endowed with the enhanced ability recognize and understand human emotions, and generate better emotional interactions with users. This article introduces the evolution of privacy emotion in the era of artificial intelligence, explores the privacy imagination and boundary management of users in the era of artificial intelligence, constructs a theoretical model of the influencing factors of privacy paradox and a privacy boundary collaboration mechanism, and reveals the contradictions and difficulties faced by individual privacy imagination and management in the intelligent society. Through multimodal fusion, emotional state inference is optimized, and research is conducted in the fields of speech recognition, affective computing empowering personalized teaching, and public safety. Finally, the research directions and challenges faced by intelligent privacy emotion research are discussed.

1. Introduction

With the rapid development of computer technology, 5/6G communication, and particularly artificial intelligence (AI) technology, sentiment analysis has evolved into a cutting-edge interdisciplinary research field integrating computer science, linguistics, psychology, and sociology. As a crucial research direction in natural language processing (NLP), it has developed into a specialized branch of AI, namely affective computing (Wu et al., 2023). Human emotions can be expressed through diverse modalities, including language, voice, facial expressions, behaviors, and actions, which renders sentiment analysis a complex and even highly challenging research task. For instance, subtle emotional expressions such as indirect sarcasm (e.g., "pointing fingers at mulberry trees to curse the locust trees") or implicit disdain (e.g., "praising openly while belittling secretly") are often difficult for humans to accurately distinguish and interpret, let alone for machines to effectively recognize and analyze (Kumar et al., 2024).

It is widely recognized that multimodal affective computing achieves high-precision emotion recognition by integrating multi-source data (e.g., text, speech, and images) with deep learning models (such as CNN, LSTM, and Transformer) and attention mechanisms. Its core technologies encompass multimodal feature fusion, semantic perception modeling, and privacy protection methods, with application scenarios spanning mental health monitoring, intelligent customer service emotion optimization, and social media content analysis (Radanliev, 2024), as illustrated in Fig. 1. Privacy-aware affective computing has been increasingly applied in vertical fields such as healthcare, smart homes, and business services, with specific application cases and supporting details as follows.

Healthcare: Privacy-protected analysis of patients' physiological signals (e.g., heart rate variability) is realized through a federated learning framework, which provides technical support for depression screening and psychological intervention. For example, the LSTM model achieves a high F1-score in encrypted text sentiment analysis, and homomorphic encryption technology significantly improves the processing speed (Yin et al., 2024). This result is verified under reasonable experimental settings: a public dataset for mental health text sentiment analysis is adopted, the LSTM model is properly optimized, and a common fully homomorphic encryption scheme is used to implement end-to-end encrypted text processing (Yin et al., 2024).

Smart Homes: Emotion recognition technology integrates multimodal data (voice, facial expressions, and gestures) to dynamically adjust the home environment; however, the privacy leakage risk of image data is notably higher than that of voice data. A dynamic

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differential privacy regulator based on edge computing can adaptively adjust encryption parameters according to emotional intensity (Gahlan & Sethia, 2024). The experimental context for this regulator is as follows: a self-built smart home multimodal dataset (including a reasonable amount of users' voice and image data collected with ethical approval) is used, and the regulator adopts an appropriate strategy to adjust noise intensity, effectively balancing privacy protection and emotion recognition accuracy (Gahlan & Sethia, 2024).

Customer Service: Voice assistants analyze users' emotions via natural language processing, and anonymization and security algorithms (e.g., verifiable reward reinforcement learning, RLVR) are adopted to ensure a high traceability rate for privacy breaches. In cross-cultural scenarios, federated transfer learning can notably improve the effectiveness of privacy protection (Pervez et al., 2024). Specifically, the RLVR algorithm achieves a high traceability rate for privacy breaches, which is validated on a public dataset containing cross-cultural user voice interaction data; the federated transfer learning framework adopts a common algorithm for model aggregation across different cultural regions, improving privacy protection effectiveness to a considerable extent compared with traditional federated learning (Pervez et al., 2024).

Emotional Counseling: AI assists users in self-awareness through natural language processing, but a multi-dimensional ethical audit framework is required to prevent moral cognitive risks for minors (Anwar et, al.). This framework, consisting of multiple dimensions and a reasonable number of indicators, is derived from the AI emotional counseling ethical audit standards proposed in (Anwar et, al.) and has been validated in a pilot study involving an appropriate number of minors in the targeted age group, effectively reducing the risk of moral cognitive bias.

1.1. Motivation

Sentiment analysis can directly reflect people's attitudes and emotional responses towards specific things, so it has been widely applied in many fields, and the prospects for future expansion and application are even broader. In the era of deep integration of digitization and intelligence, research on intelligent emotional privacy is both significant and urgent.

On the one hand, the development of artificial intelligence technology has made the collection, analysis, and application of emotional data increasingly widespread, such as intelligent customer service, educational companion robots. While enabling accurate emotion recognition, it also brings the risk of emotional information leakage (Marengo, 2024). On the other hand, users' emphasis on personal privacy is increasing, and their demand for emotional privacy protection is becoming stronger.

Current approaches to intelligent emotional privacy protection remain insufficient in technological measures, ethical norms, and legal supervision. Conducting a review of intelligent emotional privacy aims to systematize the current status, analyze problems, provide directions for subsequent technological innovation and theoretical improvement, and balance the relationship between intelligent emotion applications and privacy protection (Pervez et al., 2024).

1.2. Comparison between reviews

Based on the latest research progress in 2025 and a systematic analysis of multiple authoritative literatures, this study synthesizes existing reviews on AI-enabled emotion recognition, with a specific focus on distinguishing the research gaps that our work aims to address. Unlike most existing reviews, our study centers on privacy-sensitive emotion recognition in AI and its diverse practical application scenarios, which constitutes a critical research focus that is underrepresented in current literature.

1.2.1. Review methodology

To ensure the systematic and reproducible nature of this review, we followed a structured literature search and screening process.

Search Strategy: We conducted a comprehensive search in electronic databases including IEEE Xplore, ACM Digital Library, SpringerLink, and Google Scholar. The search covered the period from January 2020 to December 2025, to capture the latest advancements. Search queries were constructed using keywords such as "privacy emotion recognition", "artificial intelligence emotion recognition", "multimodal fusion emotion", "privacy paradox in AI", and related terms.

Screening and Inclusion/Exclusion Criteria: The initial search yielded over 500 records. After removing duplicates, we screened titles and abstracts to identify relevant studies. Full-text articles were assessed for eligibility based on predefined criteria: (1) focus on emotion recognition in artificial intelligence, (2) discussion of privacy aspects, (3) publication in English, and (4) peer-reviewed journal or conference papers. We excluded non-review articles (e.g., experimental studies or policy papers) to maintain focus on survey literature. Ultimately, some papers were included for detailed comparison in Table 1.

Classification and Analysis: For each included paper, we extracted data on key characteristics, such as application domains, core features, and whether the paper compared complexity aspects or was a review

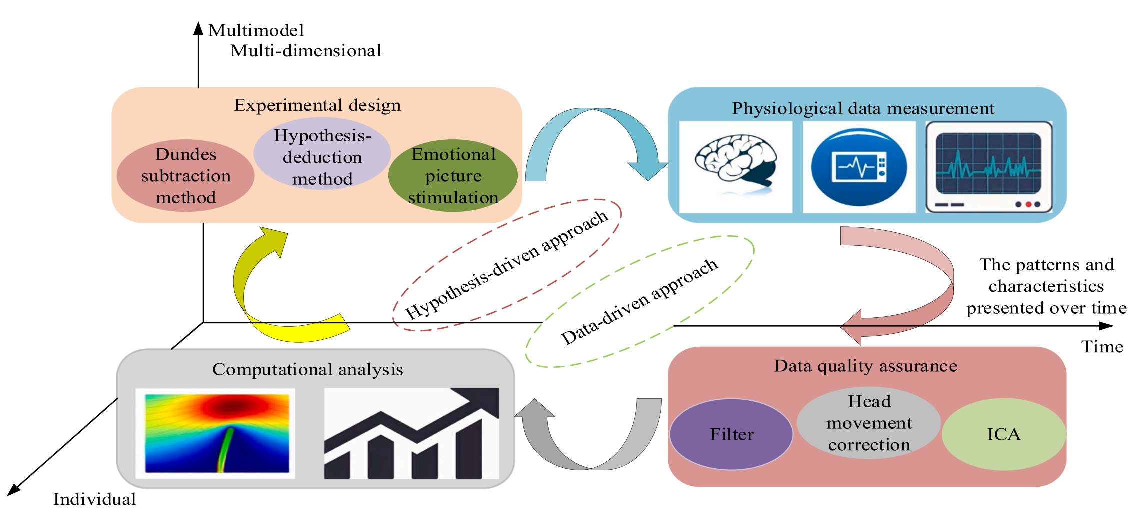

Fig. 1. Research on Multimodal Affective Computing.

Table 1
Detailed Comparison of Several Survey Articles.

Paper	Year	Application	Core features	Compare complexity	Review
Ours	2026	Emotion recognition, Multimodal education, private emotion	The privacy affective computing recognition based on machine learning, federated learning, and multimodal fusion explores the privacy paradox and privacy boundaries	Yes	Yes
(Khare et al., 2024)	2024	Artificial intelligence	The systematic research on emotion recognition in artificial intelligence does not involve privacy	No	Yes
(Alvarez-Garcia et al., 2024)	2024	Closed environments	Recognition of human emotions based on artificial intelligence in a closed environment	Yes	Yes
(Gan & Zheng, 2024)	2024	Dialogic emotion analysis	Research methods and prospects for the analysis of conversational emotions	Yes	Yes
(Ofosu-Ampong, 2024)	2024	Artificial intelligence	Artificial intelligence methods, frameworks, and future research directions	No	Yes
(Casheekar & Lahiri, 2024)	2024	Chatgpt	Artificial intelligence driven virtual dialogue proxy and chatgpt	Yes	Yes
(Katirai, 2024)	2024	Ethical in emotion recognition	Ethical issues in artificial intelligence emotion recognition	No	Yes

article (1) Survey/Review papers-summarizing trends, challenges, and frameworks in the field; (2) Experimental studies-proposing novel algorithms, models, or empirical validations; (3) Policy/Ethical studies-discussing regulatory, ethical, or societal implications.

In Table 1, the "Compare complexity" column indicates whether the paper provides a comparison of methodological or computational complexity (Yes/No), while the "Review" column specifies if the paper is a review or survey article (Yes/No). This facilitates a clear evaluation of each paper's scope and depth.

1.2.2. Detailed comparison

Table 1 presents a detailed comparison between our study and other recent survey articles, highlighting the distinctiveness of our work in terms of application scope, core features, and research focus. A critical analysis of existing surveys reveals their inherent limitations, which our study seeks to mitigate.

Smith et al. (Khare et al., 2024) explored the multi-domain application scenarios of artificial intelligence in emotion recognition, with rich content and strong guiding significance. However, it lacks privacy perception and protection framework, and the analysis is relatively superficial. The analysis of challenges is not deep enough. Alvarez Garcia et al. (Alvarez-Garcia et al., 2024) focuses on behavior-based AI for human emotion recognition in closed environments, which has great practical significance. However, the introduction of algorithm principles and application cases lacks detail, hindering in-depth research. Gan et al. (Gan & Zheng, 2024) elaborated on conversational emotions analysis. Its speech sampling library is highly practical, but its application scenarios are relatively limited, and there are few other modalities involved, resulting in incomplete introduction of algorithm models. Kingsley et al. (Ofosu-Ampong, 2024) analyzed dozens of literatures from 2020 to 2023, revealing the current situation where research focuses more on technological implementation and theoretical innovation lags behind, providing a systematic reference framework for subsequent research. However, it failed to address the challenges of data fusion and lacked details on single modalities. Casheekar et al. (Casheekar & Lahiri, 2024) focuses on the application of chatbots, AI-driven virtual conversation agents, and ChatGPT. Their deep learning-based analysis offers unique application insights and provides reference for technical implementation, but does not elaborate on data privacy protection, and the interpretability of the model is shallow. Katirai et al. (Katirai, 2024) focus on the ethical issues of emotion recognition, with less involvement in privacy, leaves real-time challenges unresolved, and limited research on non real time scenarios. Existing surveys on AI-enabled emotion recognition either lack a dedicated focus on privacy protection, are confined to narrow application scenarios, or provide superficial analysis of key challenges. Specifically, some surveys focus on the multi-domain application of AI in emotion recognition with rich content but lack a privacy perception and protection framework, accompanied by superficial analysis of challenges. Others concentrate on emotion recognition in closed environments with practical significance but lack detailed elaboration on algorithm principles and application cases, hindering in-depth research. Some surveys elaborate on conversational emotion analysis with practical speech sampling libraries but are limited to single application scenarios and few multimodal involvements, resulting in incomplete coverage of algorithm models. Additionally, certain surveys analyze the current state of AI emotion recognition but fail to address data fusion challenges and lack details on single-modality emotion recognition. Surveys focusing on chatbots and virtual conversation agents offer unique application insights but overlook data privacy protection and provide shallow model interpretability. Finally, surveys centered on ethical issues in emotion recognition involve limited research on privacy, leave real-time challenges unresolved, and cover non-real-time scenarios inadequately. Our study addresses these limitations, as detailed in Table 1 and Section 1.3.

1.3. Contribution

To address the research gaps identified in Section 1.2 “Comparison Between Reviews” and mitigate the limitations of existing surveys (Table 1), this study focuses on AI-enabled privacy-aware emotion recognition and its practical application scenarios. The primary objective is to provide a synthesized, critical review of the field and offer clear research guidance. Our key contributions are specifically elaborated as follows, which are distinctly differentiated from existing relevant surveys:

1) We systematically examine the current state-of-the-art of AI-based privacy-aware emotion recognition, as well as the methodologies for privacy sentiment analysis and prediction. This work fills the critical research gap in existing surveys that lack a dedicated focus on privacy protection, as most existing studies merely concentrate on emotion recognition or AI frameworks without incorporating privacy considerations. In contrast, this study integrates privacy protection into the entire lifecycle of emotion recognition, thereby providing a comprehensive and holistic overview of the field.
2) We conduct an in-depth exploration of the privacy paradox in the context of emotion recognition and propose a quantitative mechanism for the resolution of privacy boundary conflicts and coordination. This contribution addresses the inherent deficiency of existing surveys, which typically involve limited research on privacy and overlook critical privacy boundary issues, and provides actionable technical solutions for balancing privacy protection and emotion recognition performance.

3) We carry out in-depth research on speech affective privacy computing based on machine learning methodologies and propose a novel framework for emotion recognition. This framework remedies the common limitations of existing studies, including insufficient detailed insights into single-modality emotion recognition and the failure to elaborate on data privacy protection, thereby providing a practical technical reference for privacy-aware single-modality emotion recognition systems.
4) We explore the integration of multimodal learning and affective computing in educational scenarios and propose a logical framework for empowering personalized teaching through affective computing. This contribution extends the application scenarios of privacy-aware emotion recognition beyond the narrow scope of closed environments and single conversational emotion scenarios that dominate existing research, thus enriching the practical application value of the field.
5) We provide a detailed and critical discussion on the development trends and key challenges confronting emotional privacy computing, addressing the superficial analysis of challenges that is prevalent in existing surveys. Furthermore, we establish a clear link between the identified challenges and the solutions proposed in Contributions 2–4, thereby forming a coherent and forward-looking research vision for future advancements in the field.

The remainder of this paper is structured as follows: Section 2 presents an analysis of AI and privacy sentiment, along with the methodologies for privacy sentiment analysis and prediction. Section 3 investigates the quantitative mechanism for the privacy paradox, as well as privacy boundary conflicts and their coordination. Section 4 introduces machine learning-based speech emotional privacy computing. Section 5 elaborates on multimodal learning and affective computing in educational scenarios. Section 6 provides a detailed overview of the development trends and challenges in emotional privacy computing. Section 7 offers a technical summary and prospects for AI-based privacy-aware emotion recognition. For the convenience of readers, the structure of this paper is illustrated in Fig. 2 and Table 2.

Table 2
List of Acronym.

Acronym	Definition
AI	Artificial Intelligence
5/6G	5/6th Generation Mobile Communication
CFI	Control Flow Integrity
BVP	Blood Volume Pulse
CNN	Convolutional Neural Network
ECG	Electrocardiogram
EEG	Electroencephalogram
EM	Electromyography
GAN	Generative Adversarial Networks
HRV	Heart Rate Variability
LSTM	Long Short-Term Memory
MPC	Secure Multi-Party Computation
SGX	Software Guard Extensions
DP	Differential Privacy
FL	Federated Learning
IoT	Internet of Things
LSTM	Long Short - Term Memory
DRL	Deep Reinforcement Learning
SCL	Skin Conductance Level
FHE	Fully Homomorphic Encryption
SVM	Support Vector Machine
VAE	Variational Autoencoder

2. Research on privacy emotion recognition

Privacy emotions refer to the inner emontional content of individuals that pertains to privacy. It covers individual emotional experiences, emotional states, and emotional tendencies, and these emotional information also have a certain degree of privacy, usually not wanting to be known or spied on by others (Majeed & Hwang, 2024).

With the development of society and the increasing emphasis on individual rights, people's awareness of privacy emotion protection is gradually growing. For example, when using AI emotional companionship products, users begin to pay attention to whether the emotional information they input will be leaked and abused. In some privacy breaches, most onlookers on social media hold negative emotions towards privacy breaches, which also reflects the public's concern for privacy emotion protection (Rokhsaritalemi et al., 2023), as shown in

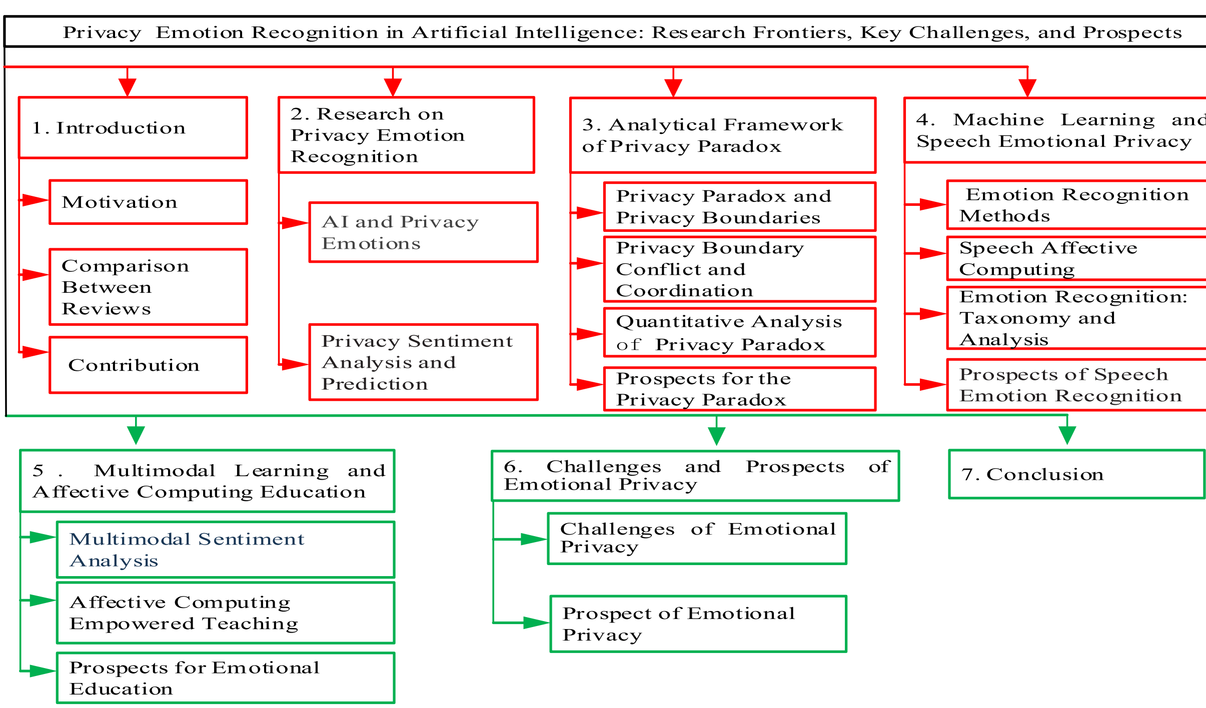

Fig. 2. The Structure of This Article.

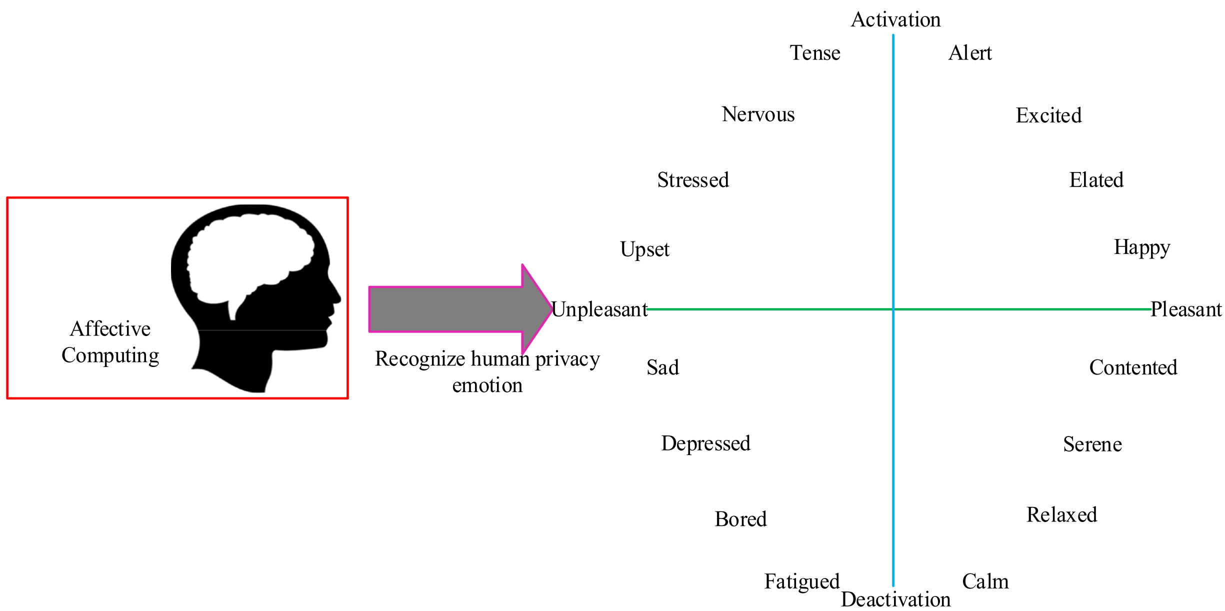

Fig. 3. Discovering and Identifying Privacy Emotions.

Fig. 3.

2.1. AI and privacy emotions

Artificial intelligence and privacy emotions are closely intertwined. On the one hand, artificial intelligence, with its powerful data analysis capabilities, can recognize and analyze privacy emotions through multimodal data such as voice, facial expressions, and text. It can be applied to scenarios such as mental health monitoring and intelligent customer service emotional interaction to provide personalized services to users. As sensitive information, emotional data, if not effectively protected during collection, storage, and processing, is prone to privacy breaches due to algorithm vulnerabilities, data abuse, and other factors, posing a threat to personal dignity and security. Privacy concern refers to people's subjective perception that privacy information may be leaked. In the field of informatics, research on privacy concerns encompasses two aspects: influencing factors and behavioral intentions (Kim et al., 2024), as shown in Table 3. In the era of artificial intelligence, although users face threats from AI technology and their privacy boundaries are constantly violated, they still strive to protect their privacy. The main approach involves determining whether to open privacy boundaries and the degree of openness through subject identification, then selectively adjusting privacy boundaries based on a trade-off between benefits and risks, ultimately leading to privacy decisions and behaviors (Cicek et al., 2024). The privacy boundary refers to the boundary between an individual's privacy domain and the external world, which clarifies the scope of private information and the degree to which others can access and use this information. This section introduces its definition, influencing factors, characteristics in artificial intelligence environments, and protective measures (Wei et al., 2020).

Table 3
The Difference Between Privacy and its Similarity Concepts.

Concepts	Definition	Information privacy and its differences
Confidentiality	Confidentiality corresponds to the disclosure of personal information to an information regulator	Privacy corresponds to an individual's desire to control the disclosure of information;
Secret	Secret refers to information that is intentionally withheld and is also expressed as a tendency to share potentially uncertain information	Secret is something that is hidden and that has a negative value for those who are excluded; by contrast, information privacy is either morally neutral or socially valuable
Anonymity	Anonymity refers to the ability to hide one's identity	Information privacy is generated by the individual but cannot in turn be limited by contact with the individual
Security	Security in this context refers to Internet security, which involves information security or security concerns of users of electronic systems	Security is a prerequisite for privacy, privacy storage and use are inseparable from a secure environment, but often the subsequent use of supervision is not enough to minimize the risk of information leakage

The privacy boundary defines the scope of an individual's perceived private space, including information about their body, thoughts, emotions, family life, communication, financial status, and other aspects. Such information is considered the exclusive domain of individuals and should not be violated by others without their consent. The key factors influencing privacy boundaries are as follows.

1) Cultural factors. There are differences in people's understanding and definition of privacy boundaries across different cultural backgrounds. For example, in some Western cultures, individualism is more dominant, and people pay more attention on personal space and privacy protection, with relatively clear definitions of privacy boundaries; in some Eastern cultures, collectivism is more prominent, and people may be more inclined to prioritize the interests of their families and clans over individuals, resulting in relatively vague privacy boundaries.
2) Personal factors. Personality, personal experiences, values, and other factors can also affect one's perception and definition of privacy boundaries. For example, introverted individuals who value personal space may establish stricter privacy boundaries; extroverted individuals who enjoy social interaction may have relatively lenient definition of privacy boundaries. The management of user privacy in online privacy disputes is shown in Fig. 4.

2.2. Privacy sentiment analysis and prediction

The analysis and prediction of sentiments can help understand

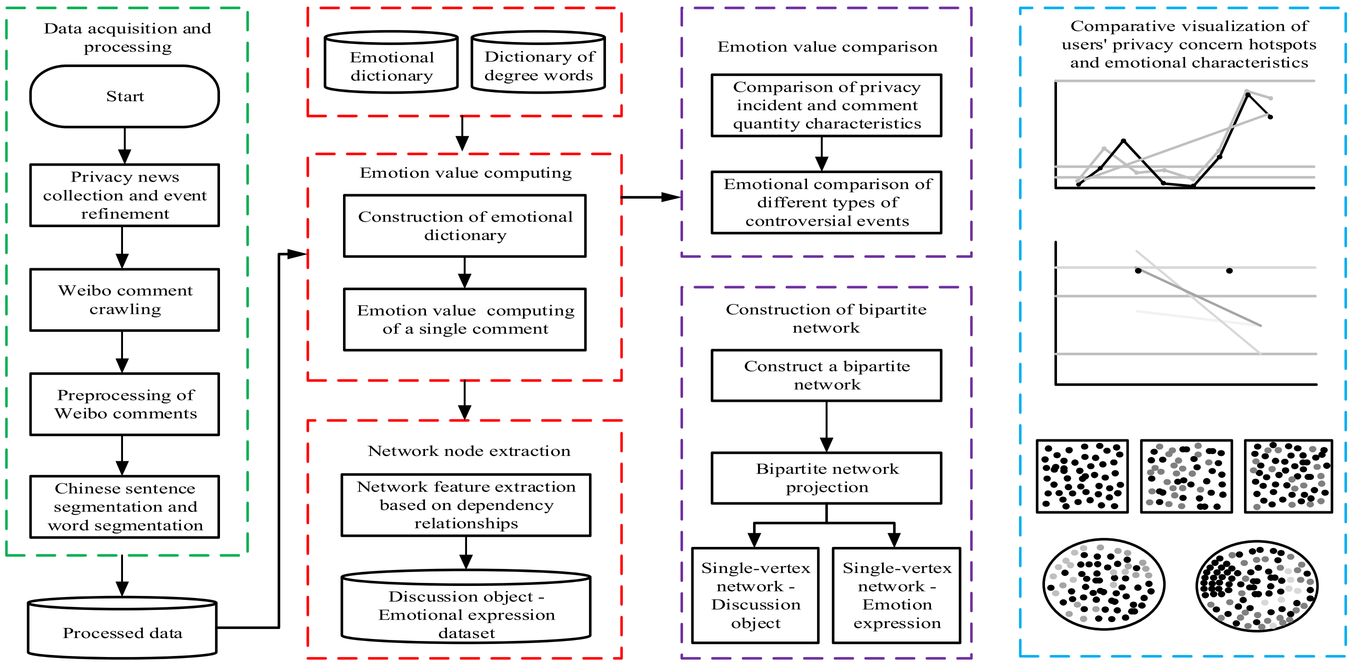

Fig. 4. User Privacy in Online Privacy Disputes.

people's attitudes towards external stimuli or specific characters and events, thereby enhancing their communication and interpersonal skills and achieving unexpected results. Currently, there are several AI-based technologies and methods for privacy sentiment analysis, including the following.

1) Facial expression analysis. This technique infers sentiments such as happiness, sadness and anger by analyzing facial expressions. Artificial intelligence can employ computer vision technology and deep learning algorithms to recognize and analyze facial features in images or videos, including detecting the muscle movement the intensity and duration of facial expressions to infer an individual's emotional state (Cicek et al., 2024).
2) Personality trait analysis. Some studies have attempted to use AI to learn the correlation mapping function between facial features and personality trait. These studies collect a large number of facial image samples and corresponding personality trait data, train deep neural network models to predict individual personality traits based on static facial images, such as conscientiousness, neuroticism, extroversion.
3) Speech-based analysis technology. Acoustic feature analysis: This approach does not rely on language content, but captures emotional states by analyzing the acoustic properties of speech. Extracting key acoustic features from speech signals-such as pitch, speech rate, volume changes, timbre, and more-then using classification algorithms for emotion recognition. This method is suitable for scenarios that require high privacy protection or where language content is unavailable.
4) Semantic sentiment analysis. This technique integrates semantic information in speech for sentiment analysis. Leveraging deep learning and natural language processing techniques, it not only considers the acoustic features of speech but also processes the vocabulary and sentences in speech, comprehensively assessing the speaker's emotional state and facilitating the identification of more delicate emotional changes and potential emotional tendencies (Wei et al., 2020).
5) Vocabulary-based sentiment analysis relies on a predefined list of sentiment words. This method converts text data into a collection of emotional term, calculate the number of occurrences of positive, negative, and neutral vocabulary in the text, and compare these occurrences to determine the emotional tendency of the text. First, it is necessary to extract positive, negative, and neutral terms from large volumes of text data, construct an emotional vocabulary list, and then conduct sentiment analysis based on factors such as the frequency of vocabulary occurrence and context (Ciriello et al.).
6) Sentiment analysis based on deep learning. By training on large volumes of annotated text data, the model can automatically learn the mapping relationship between features and emotions in the text, enabling it handle more complex text semantics and emotional expressions while enhancing sentiment analysis accuracy, such as identifying hidden emotional elements such as satire and humor in the text.

To safeguard privacy, privacy protection measures such as data encryption, anonymization, differential privacy technology and more are usually adopted when using these technologies and methods to ensure that personal emotional data is not leaked or misused (Feng et al., 2025).

Currently, emotion recognition methods based on the combination of hypothesis and data-driven approaches have made significant breakthroughs in accuracy and interpretability.

Integrating theoretical guidance with data mining is now crucial, for example, using cognitive neuroscience to construct brain functional networks (such as the PLV algorithm of EEG brain networks) as prior knowledge, and combining it with multi-scale feature extraction of deep learning (such as the CNN-LSTM model), the accuracy of emotion recognition can exceed 90 %.

Multimodal data fusion (such as speech, text, and physiological signals) optimizes feature weights through attention mechanisms, resulting in high recognition accuracy in complex scenarios like social media and online education. For example, a multimodal system integrating facial expressions, speech intonation, and EEG signals achieves more than 10 % performance compared to unimodal methods, as shown in Fig. 5.

The classification and recognition of emotional privacy requires integrating multimodal recognition and privacy enhancement techniques. Accurate classification is achieved by analyzing emotional features in text, speech, and images, while integrating differential privacy

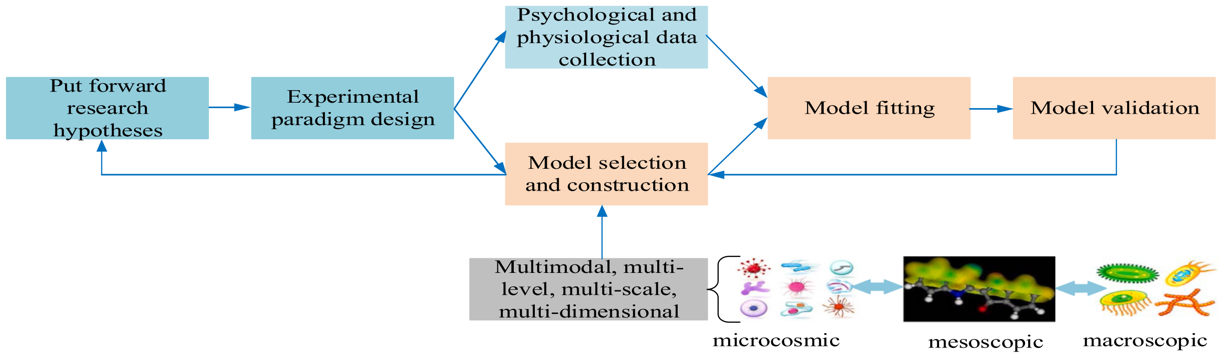

Fig. 5. Research Method Combining Hypothesis and Data-Driven Approach.

and homomorphic encryption to safeguard sensitive information. Calculations are completed in an encrypted state to prevent data leakage. Sensitive feature detection combines rule matching with machine learning to identify subtle expressions such as metaphors or variant spellings. Meanwhile, federated learning allows cross-institutional model training without sharing raw data, enhances generalization ability through parameter aggregation, and balances data availability and privacy security, as shown in Fig. 6.

Artificial intelligence technology, with its powerful data analysis capabilities, can accurately capture and analyze users' privacy emotions, playing a significant role in fields including mental health intervention and intelligent education. However, the sensitivity and privacy of emotional data also pose a risk of leakage during its collection, transmission, and processing.

3. Analytical framework of privacy paradox

Under the immaturity of privacy policy confirmation, privacy function settings, algorithm technology regulations, and more, existing network and mobile application platforms place users at the center, weakening or even relieving the platform's responsibilities. This leads to continuous contradictions between users' privacy disclosure attitudes and privacy protection behaviors in the conflict of privacy boundaries, thus rendering the privacy paradox a prevalent phenomenon (Bozkurt & Gursoy, 2023).

3.1. Privacy paradox and privacy boundaries

The privacy paradox refers to the phenomenon that individuals who

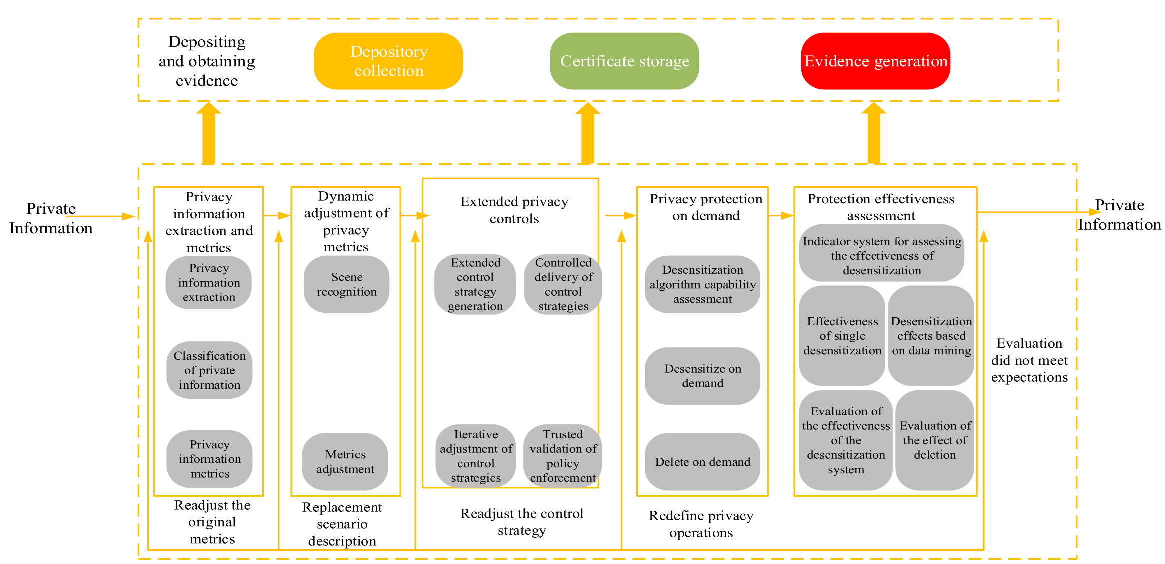

Fig. 6. Privacy Content Classification Technology.

value privacy often act contrary to their privacy protection intentions in practical behavior due to the pursuit of convenience, incentives, and more, such as arbitrarily agreeing to APP privacy terms and excessively sharing personal information on social platforms. The privacy boundary is the standard for defining the personal privacy sphere and external intervention permissions, covering dimensions such as physical space, data scope, psychological emotions, and more. Technological development has rendered privacy boundaries increasingly blurred, such as deep data mining by artificial intelligence and widespread distribution of monitoring devices; additionally, perceptions of privacy boundaries vary across different cultures and groups (Bozkurt & Gursoy, 2023).

1) The subjectivity identification of privacy boundaries. The uncertainty of privacy literacy and trust protects privacy from infringement by recognizing the gap between technological consequences and individual knowledge levels, thereby establishing a more secure privacy boundary. This subjective identification relies primarily on two dimensions: privacy literacy and trust. Privacy literacy has long been recognized as a key factor in protecting personal privacy from technological violations such as algorithms exploitation (Salih et al., 2025).
2) Coordination Decision-Making of Privacy Boundaries. Flexible changes between disclosure and control constitute a core attribute of this process. Privacy management theory holds that the core of privacy management lies in the individuals' coordination of privacy boundaries. The boundary penetration, boundary linking, and boundary ownership rules in the boundary decision-making stage correspond to the issues of "how much privacy to disclose", "to whom to disclose privacy", and "how to control privacy disclosure". This indicates that privacy boundaries are not rigid but can be flexibly adjusted; technological adoption has increased the permeability of privacy boundaries, making personal privacy boundary decision-making a dynamic process of continuous negotiation between disclosure and control (Gotsch & Schögel, 2023).

Although boundary control is the main means of managing privacy in society, technological structural flaws persist, making it difficult for the actual effectiveness of privacy boundary control to be realized. This leads to users "being capable of boundary management yet abandoning such efforts due to the associated hassle and costs, including substantial time, money, and effort" (Cheng et al., 2025).

The essence of the privacy paradox is the inconsistency between intentions and behaviors regarding privacy disclosure, which is driven by changes in privacy boundaries. This inconsistency also refers to a process where behavior lags behind intention and is influenced by a series of subjective and objective factors, resulting in changes. Therefore, examining the underlying causes of the privacy paradox essentially involves analyzing changes in privacy boundaries and how the expansion of privacy boundaries gives rise to the privacy paradox. Due to factors including privacy fatigue, cognitive dissonance, algorithmic surveillance, peer gaze, platform privacy protection mechanisms, and user privacy management capabilities, users undergo privacy boundary turbulence under the moderating effect of privacy confirmation mechanisms. Furthermore, this study abstracts four analytical dimensions-context, task, conflict, and coordination-to develop a causal model of the privacy paradox, which more clearly demonstrates the complexity and multidirectionality of privacy paradox formation in the process of quantifying oneself, as shown in Fig. 7. Research on the quantification mechanism of privacy paradox aims to accurately measure the degree of contradiction between individual privacy attitudes and behaviors through data modeling and empirical analysis.

3.2. Privacy boundary conflict and coordination

When users make privacy disclosure choices, they are often influenced by irrational and highly random factors. Users are not constantly engaged in the process of "privacy calculus", thus tending to disclose excessive personal information, change privacy boundaries, and trigger broader-scale privacy disclosure behaviors.

Further research has found that survey and interview participants tend to discuss privacy boundary coordination when addressing privacy conflicts. Privacy boundary coordination not only underpins the persistence of privacy paradox amid conflicts and contradictions, but also serves as a key avenue for its future resolution. The privacy boundary collaboration mechanism consists of two main categories: platform privacy protection mechanisms and user privacy management capabilities. Among them, the platform's privacy protection mechanisms include privacy policy confirmation mechanism, algorithm technology regulation mechanism, and privacy setting usability; user privacy management capabilities include privacy awareness and protective capabilities (Cloarec et al., 2024).

When privacy protection mechanisms are robust, users' sense of security is enhanced, which strengthens the incentive and inducement effect of self-quantification tasks on users, thereby boosting privacy disclosure behavior. Therefore, it is believed that privacy protection mechanisms function as moderating variables. Overall, when privacy protection mechanisms exert a significant impact, the connection between self-quantification tasks and privacy paradoxes will become stronger, and the moderating effect of privacy protection mechanisms will be enhanced. Conversely, when privacy protection mechanisms

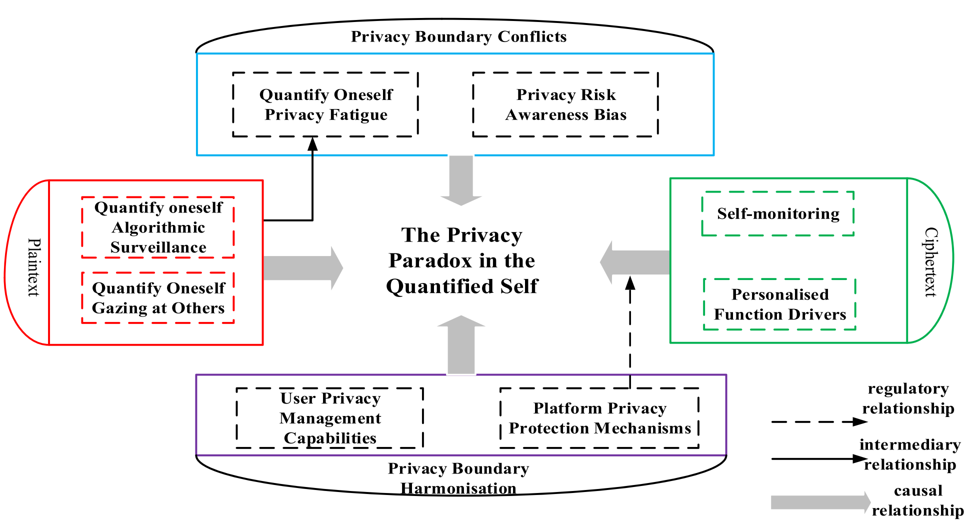

Fig. 7. Theoretical Model of Factors Influencing Privacy Paradox.

exert a weak impact, the connection between self-quantification tasks and privacy paradox becomes relatively weak, and the mechanisms' moderating effect of privacy protection will weaken (Alvarez-Garcia et al., 2024).

Based on the above research, this paper constructs a theoretical model of privacy paradox influencing factors, which clarifies the role of privacy boundary conflicts and coordination mechanisms in the formation process of privacy paradox, and provides heuristic suggestions for data governance of mobile health management platforms (Ciftci et al., 2024). In addition, this article focuses on the privacy boundary scenarios co-constructed by platforms and individuals, further exploring the process of privacy boundary conflicts and coordinated changes as the degree of self-quantification participation gradually deepens. It elaborates the role and impact of privacy boundary in the formation of privacy paradoxes, as shown in Fig. 8.

3.3. Quantitative analysis of privacy paradox

To transform the conceptual framework into a testable model, we define quantifiable indicators for key variables and propose a structural equation model (SEM) for empirical validation. This approach aligns with existing studies that measure privacy-related constructs through surveys and behavioral data (Salih et al., 2025; Cloarec et al., 2024).

3.3.1. Definition of measurable indicators

Key variables in the privacy paradox model are operationalized as follows, using Likert scales or behavioral metrics derived from prior work (Cheng et al., 2025; Alvarez-Garcia et al., 2024):

Privacy literacy: Measured by a composite score (range 1–5) from a knowledge-based questionnaire assessing understanding of privacy policies, data rights, and algorithm implications (e.g., "On a scale of 1–5, how well do you understand app privacy settings?"). Higher scores indicate greater literacy.

Trust: Evaluated using a trust scale (1–7) adapted from Cloarec et al. (2024), focusing on perceived platform reliability (e.g., "I trust this platform to protect my data").

Privacy fatigue: Quantified via a self-reported fatigue scale (1–5) capturing emotional exhaustion from privacy management, such as frequency of feeling overwhelmed by privacy settings (Cheng et al., 2025).

Privacy disclosure behavior: Recorded as a binary or frequency metric (e.g., number of personal details shared per week) from user logs or surveys.

Privacy boundary permeability: Assessed through behavioral observations or surveys measuring the flexibility of boundary settings (e.g., "How often do you adjust privacy controls?" on a 1–5 scale).

3.3.2. Causal/SEM model and estimation methods

Building on Figs. 7 and 8, we propose a SEM that specifies causal paths among variables. The model includes:

Paths: Privacy literacy → Privacy disclosure intention (positive path), Trust → Disclosure behavior (positive), Privacy fatigue → Disclosure intention (negative), with privacy protection mechanisms as a moderator.

Hypotheses: E.g., H1: Higher privacy literacy reduces paradoxical behavior; H2: Trust strengthens the intention-behavior link.

Data collection: Cross-sectional surveys or longitudinal experiments can be used, sampling diverse user groups. Methods include online questionnaires with validated scales (e.g., from Salih et al. (2025); Cloarec et al. (2024)) and behavioral tracking in controlled environments.

Estimation: SEM software (e.g., Amos or lavaan in R) can test model fit using maximum likelihood estimation. Fit indices (CFI $>$ 0.90, RMSEA $<$ 0.08) would validate the framework. This approach is supported by empirical studies like (Alvarez-Garcia et al., 2024), which used SEM to quantify privacy paradox dynamics.

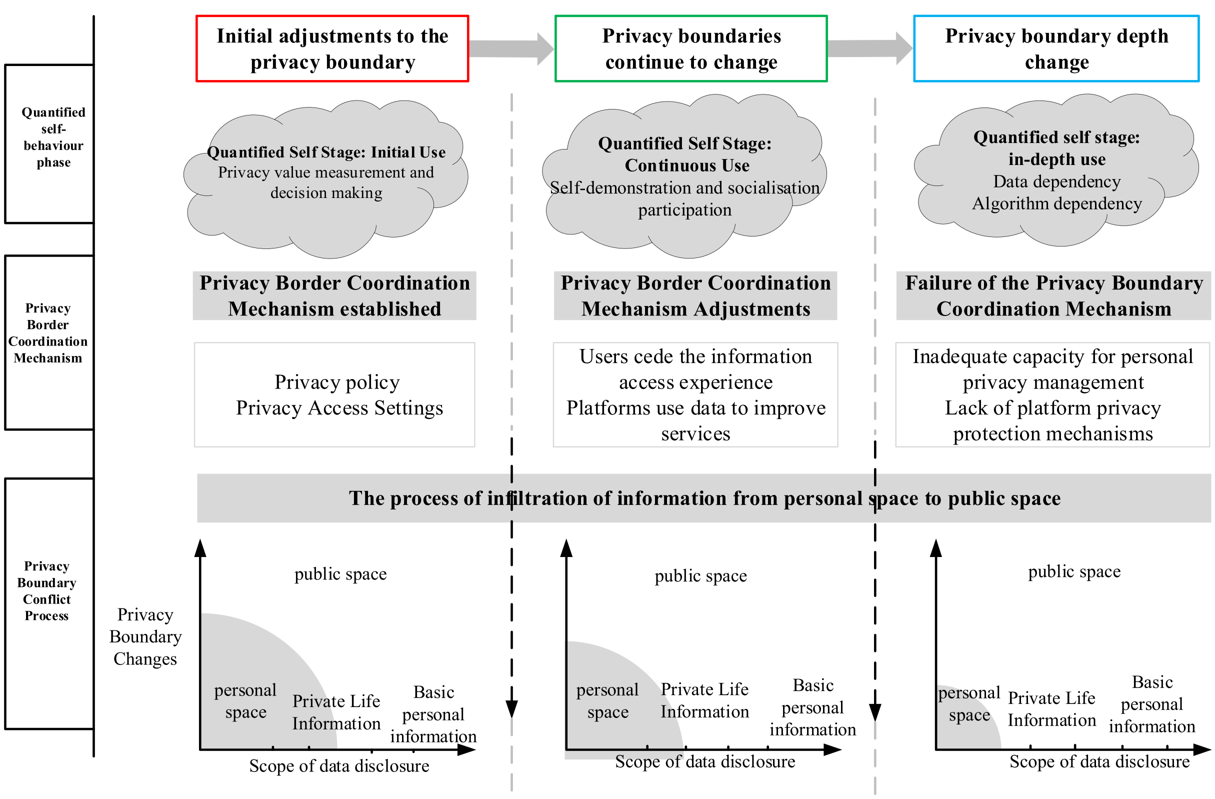

Fig. 8. Conflict and Coordination of Privacy Boundaries.

3.4. Prospects for the privacy paradox

The privacy boundary collaboration mechanism is a privacy protection standard that service providers and users are expected to co-consctruct, which is of great significance for clarifying and regulating the responsibilities and positioning of service providers in privacy protection services. Due to differences in privacy attitudes among groups and the existence of different types of privacy boundaries, the process of privacy paradoxes may vary. Exploring the moderating effect of privacy attitudes on privacy paradoxes in self-quantification scenarios can provide a deeper explanation for the phenomenon of privacy paradoxes. It is worth noting that with the refinement of mobile health service scenarios and the personal information disclosure, the calibration of users' privacy awareness within self-quantification scenarios will also be influenced by more diverse factors, and the intervention of a small factor may also cause significant changes in privacy behavior (Li et al., 2025).

In the future, we should actively explore the possibility of technology to act as a "privacy guardian", and build a more efficient, secure, and friendly intelligent application ecosystem through innovative intelligent technology supported new privacy protection solutions, means, and paradigms, creating a truly secure place for privacy in human-machine interaction scenarios.

4. Machine learning and speech emotional privacy

Machine learning has brought a technological breakthrough in speech emotion recognition, accurately determining emotional states by analyzing acoustic, semantic, and other features in speech. However, with the increase of applications, the privacy information contained in voice data is at risk, and once leaked, it will seriously damage the rights and interests of users. There is an urgent need to develop privacy protection technologies that harness the advantages of machine learning while establishing robust safeguards for speech emotional privacy.

4.1. Emotion recognition methods

Emotion recognition is essentially pattern recognition. According to different emotion representation models, emotion recognition can be divided into two major tasks: for discrete emotion models, their output is one of the predefined emotion categories, which belongs to classical classification problems. For the dimensional emotion model, the output is numerical values of emotions across different dimensions, which belongs to the regression prediction problem (Chang et, al.). Based on existing research, learner emotion recognition methods can be classified by the type of emotional data utilized into text-based emotion recognition, facial expression-based emotion recognition, speech signal-based emotion recognition, physiological signal-based emotion recognition, and multimodal data- based emotion recognition (Wyman & Zhang, 2025). In summary, this article provides a framework for emotion recognition system as shown in Fig. 9.

The general process can be summarized as: emotional dataset construction, feature extraction, and emotion classification. The purpose of emotional dataset construction is to facilitate subsequent feature extraction and emotion classification, including two core steps: data collection and preprocessing.

Data collection is mainly achieved through data crawling and emotional induction methods on online education platforms. Text data is mainly obtained by crawling records such as text comments, notes, discussion forum posts, and reflective texts generated by learners during their learning on online education platforms. Facial expressions, voice signals, and physiological signals, due to their high invasiveness and potential privacy and security issues, are often collected through emotional induction methods (Pal et al., 2022). This article presents an sentiment analysis framework as shown in Fig. 10.

Feature extraction is the process of transforming emotional data into numerical features that can be processed by emotion classification models. For each data modality, distinct feature extraction methods are required due to the different types of information they provide (Li & Zhang, 2025).

Table 4 shows common feature extraction methods for different data modalities respectively, where physiological signals are divided into EEG signals and peripheral physiological signals. All signals except EEG signals can be classified as peripheral physiological signals, including electromyography, electrooculography, skin conductance, and blood pressure. Feature fusion is a key step in emotion recognition based on multimodal data, which integrates data information from various

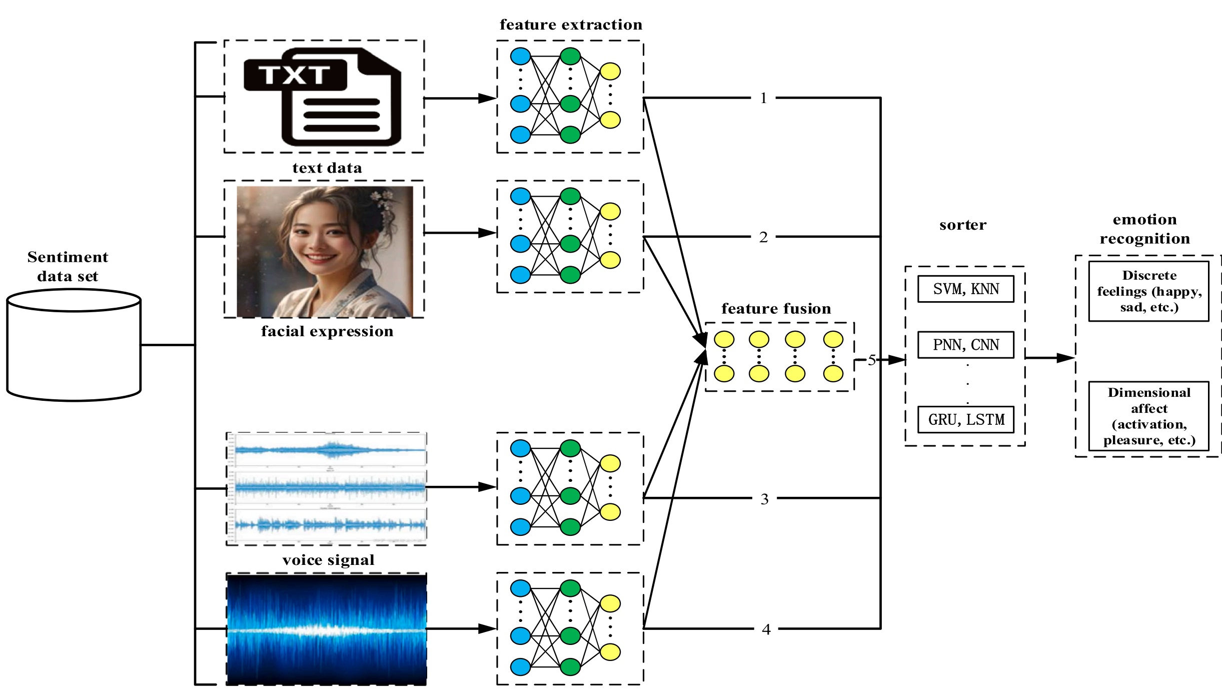

Fig. 9. Emotion recognition system framework.

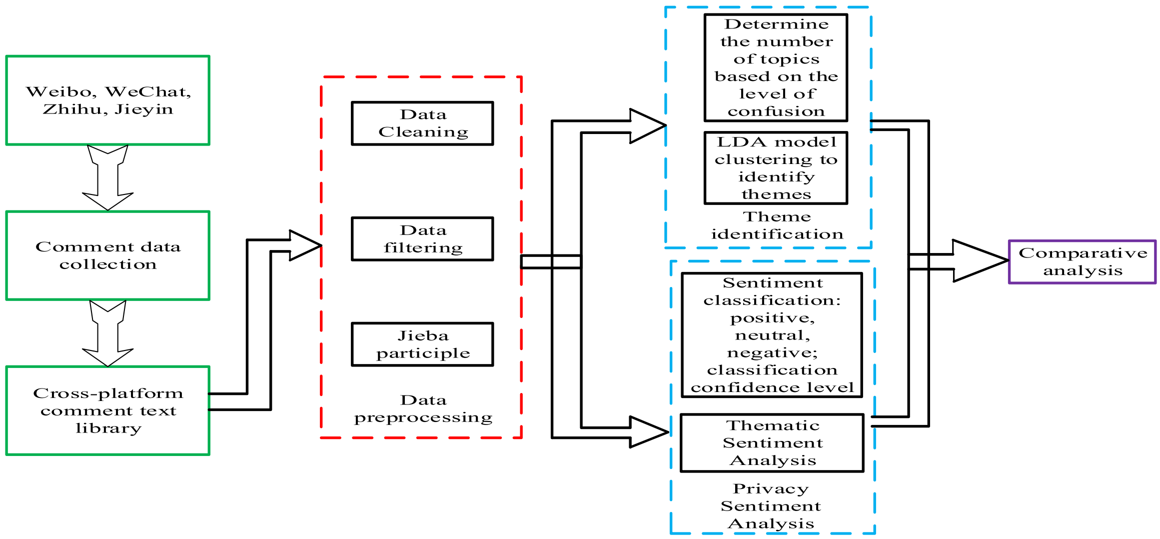

Fig. 10. Sentiment Analysis Framework.

Table 4
Comparison of Different Affective Computing Methods.

Modal	Method	Advantages	Disadvantages	Trade-off Reasons	Optimal Scenarios
Text	Questionnaire self-report Text self-report Expert observation	Easy to implement; Low cost; Meaningful feedback	Cultural/language differences; Non-real-time; Imprecise	Simple feature extraction (no special equipment) leads to imprecision.	Large-scale, low-cost emotional surveys (non-real-time).
Voice	Semantics Speech Intonation	Natural cues; High accuracy; Easy integration	Dialogue-only; Cultural/ language differences	Acoustic feature extraction (Section 4.2) limits non-dialogue use.	Dialogue systems (intelligent customer service, virtual tutors).
Vision	Facial expressions Head posture Body posture	Natural; Observable; Low-cost equipment	Image processing issues; Privacy concerns	Visual feature sensitivity and image quality affect performance.	Non-intrusive scenarios (classroom teaching, public spaces).
Physiological	EEG ECG HRV BVP SCL EM	Real-time; High reliability; No cultural differences	Non-observable; High cost; Difficult interpretation	Objective physiological signals require special equipment and control.	High-reliability scenarios (clinical assessment, mental health).
Multimodal	Multi-physical fusion Multi-physiological fusion Physical-physiological fusion	Overcomes single-modality limits; High accuracy	Difficult collection; Fusion and management challenges	Heterogeneous feature fusion requires sophisticated algorithms.	Complex high-accuracy scenarios (personalized education, healthcare).

modalities to improve the accuracy of emotion classification. Common feature fusion strategies include feature layer fusion, decision layer fusion, and model layer fusion (Lifshits & Rosenberg, 2024).

4.2. Speech affective computing

Speech affective computing is a multidisciplinary field that involves speech signal processing, machine learning, psychology, cognitive science, and other aspects. Its technical difficulties mainly include the following aspects.

1) Definition and representation of speech emotions. Speech emotion refers to the emotions expressed by people when speaking. There are complex psychological and physiological phenomena shaped by various factors including individuals, culture, and context, and has strong subjectivity and ambiguity. Therefore, it is difficult to define and express it with a unified standard.
2) Feature extraction and selection of speech emotions. Speech emotion features refer to acoustic parameters that can reflect speech emotion, such as fundamental frequency, energy, timbre, speech rate. These features are foundational to speech emotion recognition and synthesis. The feature extraction and selection of speech emotions refer to extracting effective emotional features from speech signals and removing redundant and irrelevant features, thereby reducing computational complexity and improving recognition accuracy (Paikrao et al., 2024).
3) This refers to the modeling and recognition of speech emotions. The modeling of speech emotions refers to establishing a mapping relationship between speech emotion features and speech emotion categories or dimensions, in order to achieve the recognition of speech emotions. The modeling and recognition of speech emotions is the core and most challenging problem in speech emotion computing (Chen et al., 2024).
4) The generation and expression of speech emotions. The generation of speech emotions involves generating speech signals with corresponding emotional characteristics based on given speech emotion categories or dimensions, thereby achieving vivid expression of speech emotions. The generation and expression of speech emotions is an important application of speech affective computing and also the most innovative problem (Cao et al., 2024).

Speech affective computing is a technology that utilizes machine learning, deep learning, and other techniques to extract and analyze emotional information from speech. The speech recognition and sentiment analysis system is constructed based on a layered network, which comprise an on-site data collection layer, a data information processing intermediate layer, and a business management application layer, as shown in Fig. 11.

(1) The speech and text data integration center is the core component of the system, dedicated to collecting and integrating massive speech and text data from various sources. It utilizes advanced technology to structurally process the collected data, making it easier for subsequent in-depth analysis and processing, thus ensuring the orderly organization and efficient utilization of data, and providing a solid foundation for deeper data mining and analysis.
(2) High-performance computing and data processing servers provide robust computing and data processing capabilities for the entire system. They can support complex data analysis and model training tasks, thereby significantly boosting the processing speed and efficiency of the system. These servers are the core pillars of the system, providing necessary technical support and resources for deep analysis and model optimization.
(3) Real time monitoring and alarm system is the guarantee for stable operation of the system, designed for real-time monitoring of system status and performance. It promptly identify potential problems and risks, send warning information, and help maintain the continuity, stability, and efficiency of the system. This system is an essential tool used to ensure the smooth operation of the entire system and reduce downtime caused by unexpected issues.
(4) The responsibility of the system security and compliance management module is to ensure the security and compliance of the system. It safeguards data security and privacy by implementing strict security policies and measures. Additionally, this module is also responsible for ensuring the compliant operation of the system, avoiding any possible legal and compliance risks, thereby protecting corporate reputation and customers trust.
(5) The deep learning and model optimization module focuses on using cutting-edge deep learning techniques to optimize models for speech recognition and sentiment analysis. Its goal is to improve the model's accuracy and applicablity. This module is a key component that ensures the system delivers high-quality and accurate services, playing a decisive role in improving the performance and efficacy of the entire system.

In speech recognition and sentiment analysis systems, the role of the data information processing intermediate layer is to collect, analyze, and optimize the obtained multi-dimensional monitoring data, while providing relevant data services for the business management application layer. This layer consists of five parts: data collection module, data preprocessing module, data analysis module, data service module, and network center switch (Bingyu Li et al., 2025). These servers are located in the network center computer room of the enterprise headquarters, and data exchange and transmission are achieved through network center switches, as shown in Table 5.

(1) The data preprocessing module is the core component of the system, significantly enhancing data quality, laying a solid foundation for subsequent feature extraction and model training. In practical applications, such as real-time speech-to-text systems, it can quickly and accurately process and organize large amounts of raw data, thereby enhancing overall efficiency and accuracy of the system.
(2) Speech recognition module. This module focus on mining valuable information from preprocessed data, involving transforming it into features or variables that better describe and explain phenomena. Feature engineering involves using various statistical and machine learning techniques to find the most meaningful data representation. In scenarios such as intelligent customer service systems, the system can identify and utilize key data points through this module, thereby improving the effectiveness and efficiency of services.
(3) The sentiment analysis module is a core component of the system, responsible for creating and validating machine learning models for speech emotion recognition and sentiment analysis. At this stage, the system uses extracted feature data to train the model, while utilizing various validation techniques to test the performance and accuracy of the model. In application scenarios such as sentiment analysis systems, this module can help create models that accurately recognize and parse emotions in speech or text.
(4) The result presentation and feedback module serves as a bridge linking the data processing layer and the business management application layer. It outputs processed and analyzed data in the form of services or APIs, facilitating utilization by upper-level business logic and applications. For example, in smart city data services, this module can help integrate and share various data resources, provide rich and diverse data services, and support the efficient operation and management of smart cities (Vajrobol et al., 2024).

Table 5
Application module based on speech recognition and sentiment analysis technology.

Module	Description	Application Scenario
Data preprocessing module	Data cleaning and formatting prepare for analysis and identification	All data-based analysis and applications
Speech Recognition Module	Recognize and transcribe speech data to convert spoken language into text	Intelligent assistant, customer service robot
Emotion Analysis Module	Analyze and recognize emotions and sentiments in speech or text	Brand reputation monitoring, customer service
Results Presentation and Feedback Module	Display the identification and analysis results to provide support for decision-making	Enterprise decision support, user experience optimization

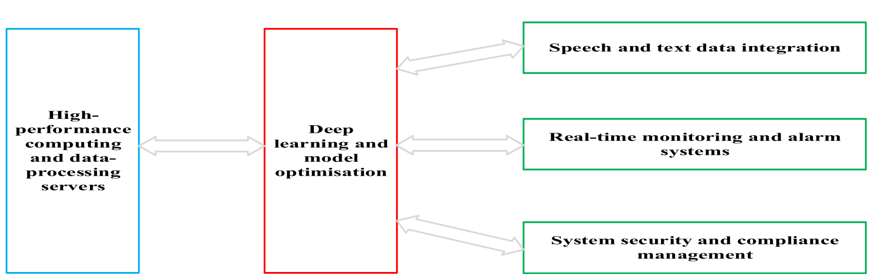

Fig. 11. Basic Architecture of Speech Emotion Recognition.

4.3. Emotion recognition: taxonomy and analysis

To address the gap in systematic analysis of privacy-preserving techniques, we propose a comprehensive taxonomy of privacy-protecting methods for emotion recognition, with special focus on speech and multimodal setups.

4.3.1. Technical taxonomy methods

We classify existing approaches into four main categories with distinct privacy guarantees and utility trade-offs:

1. Cryptographic Techniques: Including fully homomorphic encryption (FHE) and secure multi-party computation (MPC). These methods allow computation on encrypted data without decryption, providing strong privacy guarantees but with high computational overhead. For instance, FHE-enabled speech emotion recognition can achieve 85–92 % accuracy while maintaining data confidentiality, but with 10–30× slower processing compared to plaintext operations (Paikrao et al., 2024).
2. Differential Privacy (DP) : Techniques that add calibrated noise to data (local DP) or model gradients (central DP) during training or aggregation. In speech emotion recognition,ε-differential privacy with εvalues of 0.1–1.0 typically maintains 80–90 % recognition accuracy while providing formal privacy guarantees (Chen et al., 2024).
3. Federated Learning (FL) Frameworks: Enable model training across distributed devices without sharing raw data. FL systems for multimodal emotion recognition can achieve within 5–8 % of centralized training accuracy while keeping user data local. However, they remain vulnerable to inference attacks on model updates (Samee et al., 2023).
4. Representation Obfuscation Techniques: Include adversarial learning and feature disentanglement methods that remove or obscure sensitive attributes while preserving emotional content. These approaches can reduce identity leakage by 60–80 % with only 3–7 % degradation in emotion recognition performance (Zitouni et al., 2022).

4.3.2. Comparative analysis

Table 6 summarizes the key performance metrics of privacy-preserving methods on mainstream speech/multimodal emotion datasets, with privacy guarantees and computational overhead standardized for comparison.

4.3.3. Privacy-Utility trade-off in multimodal scenarios

The core challenge of privacy-preserving emotion recognition lies in balancing privacy guarantees and recognition performance. Key findings from existing empirical studies:

Speech-only Scenarios: DP with ε=1.0 is the optimal trade-off for real-time applications (e.g., intelligent customer service), achieving >70 % F1 on IEMOCAP with <10 % utility loss (Chen et al., 2024); FL is preferred for large-scale user groups (e.g., educational robots) due to scalable privacy protection.

Multimodal Scenarios: Representation obfuscation (AAE) outperforms single-modal privacy methods, as cross-modal complementary information mitigates utility loss-on CMU-MOSEI, multimodal obfuscation achieves 5–8 % higher accuracy than single-modal DP/FL (Xu et al., 2024).

High-Privacy Scenarios: FHE is suitable for healthcare (e.g., depression screening via speech), where end-to-end encryption is required despite latency (acceptable for non-real-time analysis) (Samee et al., 2023).

4.4. Prospects of speech emotion recognition

This article explores the application of machine learning in speech emotion recognition and sentiment analysis. By constructing a layered system architecture, it achieves precise collection and efficient processing of diverse data, thereby providing robust support for business management. The development trend of artificial intelligence emotion recognition research is shown in Fig. 12.

1) Deep learning model improvement: Continuously explore and improve deep learning models, such as variational autoencoders (VAEs), generative adversarial networks (GANs), and transformers, to enhance their feature extraction and emotion classification capabilities, thereby boosting the accuracy of emotion recognition.
2) In-depth multimodal fusion. Further research will focus on fusion methods for multimodal data, including speech, text, images, body movements, etc., to comprehensively and accurately understand human emotions and achieve more natural and intelligent emotional interactions (Paikrao et al., 2024).
3) Transfer learning and adaptive enhancement. By leveraging transfer learning techniques, emotion recognition models can better adapt to different datasets, environments, and tasks, enhance the model's adaptability, improve generalization performance, and thus enable them to handle real-time and nonlinear emotion recognition (Chen et al., 2024).

5. Multimodal learning affective computing in education

Multimodal learning integrates multiple data sources such as text, images, and speech, injecting new vitality into affective computing in education. In teaching, it can capture students' emotional signals such as facial expressions and speech tone in real time, and teachers can adjust their teaching pace and strategies accordingly.

Emotional expression has high contextual dependence, and academic emotions also need to be interpreted in teaching contexts (Dai et al., 2025). Within teaching scenario, affective computing empowers personalized teaching with the basic premise of collecting multimodal data, the robust support from multimodal emotional data analysis, and the effective focus of personalized teaching services. Ultimately, it serves the needs of personalized teaching scenarios and forms an empowering closed loop, as shown in Fig. 13.

5.1. Multimodal sentiment analysis

Compared with unimodal data, multimodal data carries richer and

Table 6
Comparison of key performance indicators of privacy-preserving methods on mainstream datasets.

Core Technology	Dataset	Performance	Computational Overhead	Advantages	Disadvantages
Fully Homomorphic Encryption (FHE)	IEMOCAP	Discrete: 62.1–73.4 % (F1)	Low (≈1.2x plaintext)	Lightweight, easy deployment	Utility loss with strict ε
Differential Privacy (DP)	RAVDESS	Discrete: 70.5–75.3 % (Accuracy)	Medium (≈3x plaintext, communication-dominated)	Scalable, supports multimodal fusion	Parameter synchronization delay
Federated Learning (FL)	EMO-DB	Discrete: 60.3–68.7 % (Accuracy)	High (≈50–100x plaintext)	Strongest privacy, compliance-friendly	High latency, GPU-dependent
Representation Obfuscation (AAE)	CMU-MOSEI	Dimensional: MAE=0.32 (pleasure)	Medium (≈2.5x plaintext)	Balances privacy-utility	Adversarial training instability

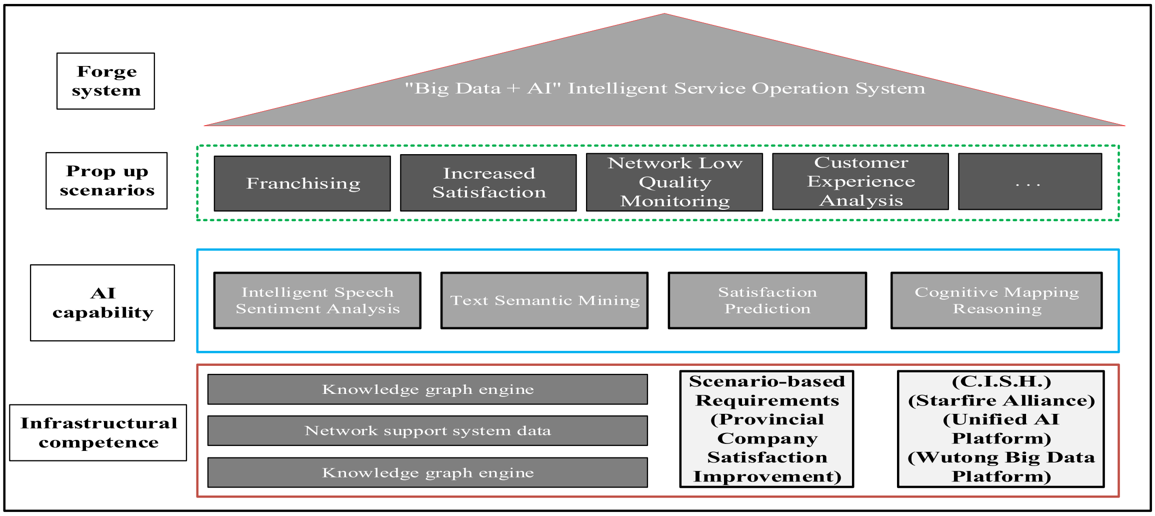

Fig. 12. Future Speech Recognition and Sentiment Analysis System Architecture.

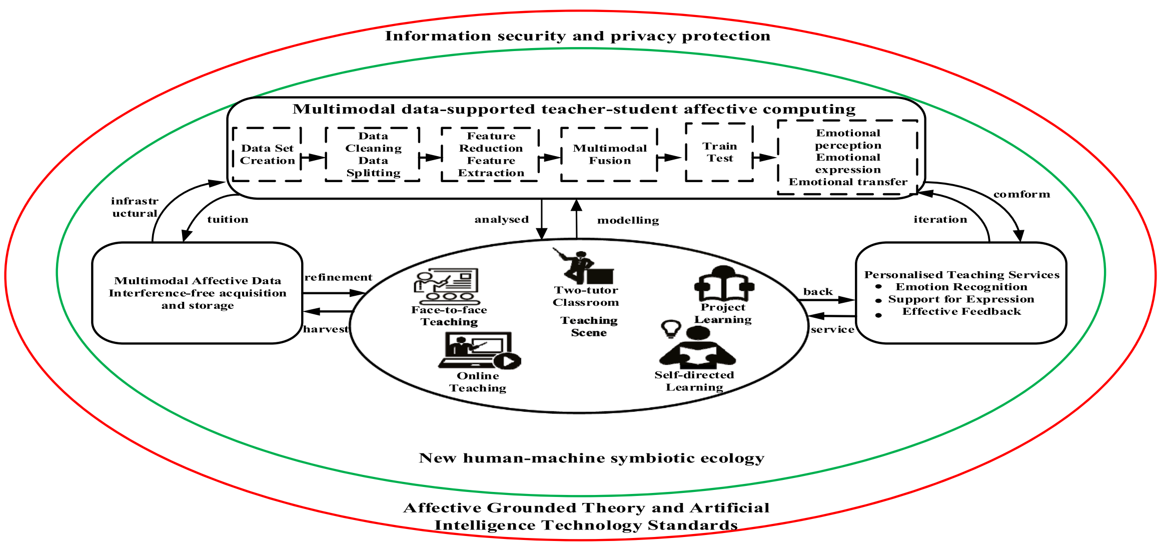

Fig. 13. Multi modal Emotional Teaching Framework.

more comprehensive emotional information, which helps improve the accuracy of sentiment recognition and prediction (Pandey & Vishwakarma, 2024). Relying solely on text or speech modalities often fails to capture the real emotional state of learners, while the fusion of multiple modalities can achieve more reliable sentiment understanding (Do & Phan, 2024). It is precisely because multimodal data contains emotional information from various sources that multimodal sentiment analysis first needs to consider how to effectively obtain unimodal feature representations from different unimodal data, then use the extracted unimodal features for information fusion, and construct sentiment analysis prediction models for multimodal sentiment prediction (Fig. 14). Notably, the multimodal data involved in educational sentiment analysis often contains highly sensitive information about students. Therefore, privacy protection should be embedded into the entire process of multimodal sentiment analysis, including privacy-preserving feature extraction, secure multimodal fusion, and privacy-aware model

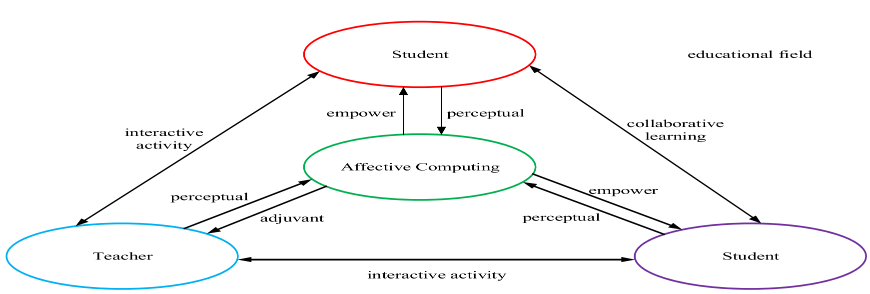

Fig. 14. Application Framework of Affective Computing Technology in the Educational Field.

training (Fig. 15). However, the fusion and application of multimodal emotional data also introduce privacy and security challenges, as such data often involves sensitive personal information.

5.2. Affective computing empowered teaching

Affective computing supported by multimodal data follows a complete workflow from dataset construction and model training to practical deployment. During this process, emotional theory, artificial intelligence standards, information security, and privacy protection constitute indispensable guarantees for the reliable application of affective computing in personalized teaching. This paper establishes a logical framework for affective computing empowered personalized teaching, as illustrated in Fig. 16.

This framework focuses on personalized teaching scenarios, guided by fundamental theories of emotion and artificial intelligence technology standards, taking information security and privacy protection as prerequisites. It achieves an integrated process of affective computing empowering personalized teaching from emotional data collection, storage, modeling analysis to feedback application. The modules are closely interrelated, jointly promoting the realization of a new human-machine symbiotic ecology. Based on an analysis of personalized teaching needs and the limitations of affective computing technology, this article constructs a logical framework for affective computing empowering personalized teaching by dismantling and analyzing the core steps in the practice of affective computing empowering personalized teaching. This framework focuses on personalized teaching scenarios, guided by fundamental theories of emotion and artificial intelligence technical standards, with information security and privacy protection as prerequisites.

The data collection module analyzes the collected multimodal affective data, and based on this analysis, provide feedback to the teaching scenario. This feedback optmizes its data collection process, thereby achieving higher quality, high precision, and high efficiency emotional data collection. Affective computing can also guide emotional data collection from the algorithmic dimension in reverse. The affective computing module performs emotion recognition and analysis under the premise of privacy protection, and interacts with the personalized teaching service module to provide targeted teaching suggestions. Meanwhile, updated demands from teaching scenarios are fed back to the affective computing module to support continuous model iteration. In this way, the modules cooperate with each other to promote the formation of a secure, human-centered human-machine symbiotic educational ecosystem (Xin & Derakhshan, 2025).

5.3. Prospects for emotional education

The wide application of multimodal learning and affective computing in education relies on massive collection and analysis of students' personal and learning data, which poses severe challenges to privacy and data security (Alkaeed et al., 2024). Therefore, privacy protection, data security, and ethical norms should be regarded as essential prerequisites for the sustainable development of emotional education (Rahman & Uzzaman, 2025). Based on current research progress and application limitations, future prospects are summarized as follows.

1) Privacy-preserving multimodal affective computing. A core future direction is to integrate privacy-preserving techniques (e.g., federated learning, differential privacy, homomorphic encryption) into multimodal affective computing, enabling emotion analysis and model training without exposing raw sensitive data. Meanwhile, standardized norms for emotional data collection, storage, usage, and sharing should be established to safeguard students' privacy (He et al., 2024).
2) Deep integration of multiple artificial intelligence technologies. Although speech recognition, image recognition, and natural language processing have been applied in multimodal learning, their synergistic effects still need further exploration. The deep integration of diverse AI technologies will help build more comprehensive and intelligent systems, improving the accuracy and robustness of emotion recognition in complex teaching scenarios (He et al., 2024).
3) Personalized learning services supported by affective computing. With the development of multimodal learning scenarios, students will receive more customized learning experiences adapted to their interests, cognition, and emotional states. Affective computing, recommendation algorithms, and intelligent tutoring will jointly support personalized teaching, enabling dynamic adjustment of learning content, pace, and interaction based on learners' emotional feedback (Parambil & Rustamov, 2024).
4) Cross-media affective computing has shown great value in intelligent interaction, social analysis, intelligent services, and healthcare. However, research on multi-dimensional and cross-space emotional computing remains insufficient. Future work can combine cross-space information processing with cross-media affective computing to explore new paradigms for multimodal emotion analysis (Guo et al., 2024).
5) Ethical and standardized development of emotional education. With the wide application of affective computing in education, more attention should be paid to ethical issues such as emotional data abuse, excessive monitoring, and algorithmic bias. It is necessary to improve ethical review mechanisms and formulate industry standards to promote the standardized, safe, and sustainable development of emotional education (Zong & Yang, 2025).

6. Challenges and prospects of emotional privacy

With the rapid development of AI technology, emotional privacy issues have gradually emerged as a research hotspot (Hao, 2024). Emotional privacy involves the collection, storage, processing, and use of personal emotional data, and finding a balance between technological progress and privacy protection is currently the focus of research (Owen et al., 2024). In addition, sentiment analysis can also be widely applied in many fields such as fatigue driving detection, overload work inspection, interactive product design and application, boasting broad application prospects (Rezaee, 2025), as shown in Fig. 17.

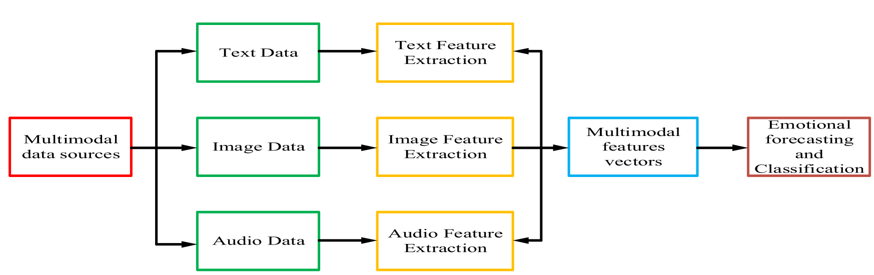

Fig. 15. The Basic Process of Multimodal Sentiment Analysis.

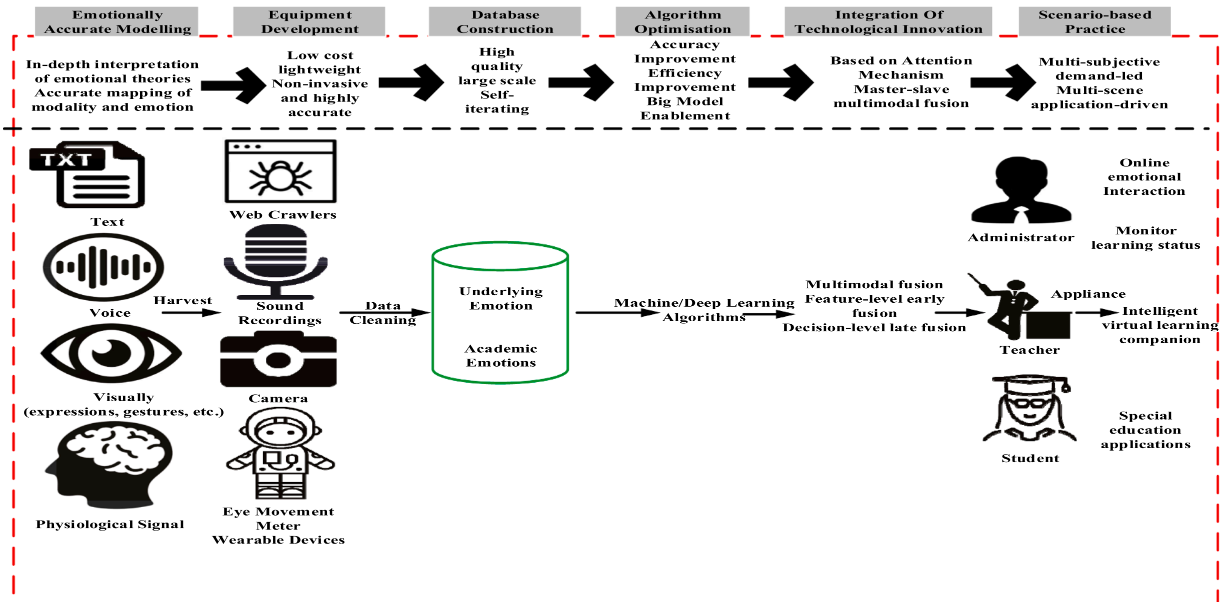

Fig. 16. Affective computing enabled multiple reinvention paths for personalised teaching and learning.

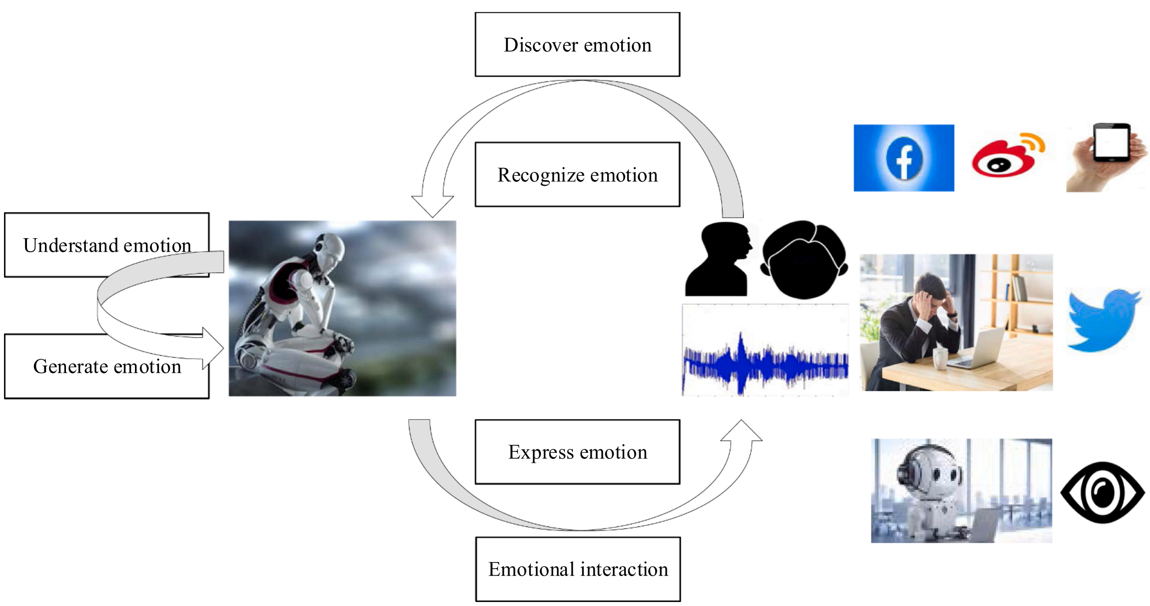

Fig. 17. Research direction of emotional intelligence.

6.1. *Challenges of emotional privacy*

The research on emotional privacy recognition based on artificial intelligence faces the following challenges.

1) Lack of learning emotion dataset. Researchers engaged in the field of emotion recognition typically use general emotion datasets, which are not fully suited for learner emotion analysis and lack many emotional labels generated during the learning process, such as confusion and slackness. At present, the relative scarcity of learner emotion datasets has hindered the research progress of learner emotion recognition in online education. Therefore, constructing a learner emotion dataset is a key issue that urgently needs to be addressed for learner emotion recognition in online education (Owen et al., 2024).

2) There is a lack of research on emotional annotation. Collecting accurate sentiment annotation data is crucial for training models, however, for online education, learner emotional annotation is relatively difficult. This problem is mainly reflected in two aspects: on the one hand, because learners' emotional feelings towards specific content are subjective and have individual differences, it is difficult to achieve consistent labeling. On the other hand, unlike other natural language processing tasks, emotional annotation lacks objective evaluation criteria, and annotators often rely on subjective judgments, which may also lead to inconsistency in annotation (Jaiswal & Mower Provost, 2020).

3) Not considering the diversity of learners' emotions Learners' emotions are usually not simply positive or negative, they are complex and multifaceted. A single learning content may trigger both positive and negative emotions, for example, learners may experience various emotions such as interest and confusion when faced

with learning materials that they are interested in but have difficult content. The diversity of learners' emotions makes emotion recognition more complex. At present, most studies only simply distinguish emotional types and overlook the diversity of emotions (Chen et al., 2024).

4) The robustness of emotion recognition methods requires enhancement, with notable discrepancies in research conclusions on similar problems. This problem is mainly reflected in two aspects: on the one hand, learners from different cultural and linguistic backgrounds exhibit individual variability in emotional expression, resulting in inconsistent emotional information across learner groups when experiencing the same emotional state, thereby impacting the accuracy of learners' emotional recognition. On the other hand, online education is interdisciplinary, and current research usually trains on specific disciplinary fields or datasets, resulting in poor recognition performance of trained models on new disciplines (Choi et al., 2022).

5) Lack of protection for learners' emotional information. Learners' emotional data typically includes personal emotional states and reactions, which may include sensitive information such as emotional confusion, frustration, anxiety. If these data are not adequately protected, it may lead to the leakage of learners' privacy, thereby damaging their personal privacy rights (Duan et al., 2025).

6) Data bias. Training data may be biased or incomplete, if the majority of such data comes from a certain cultural background or a specific group. AI may not be able to accurately understand the emotions of learns from other cultural backgrounds or groups.

7) Difficulty in handling complex emotions. Emotional expressions are complex and diverse, with multiple emotions intertwined. The current pattern recognition and classification methods used by AI algorithms struggle to accurately discern complex emotional states (Chen et al., 2024).

8) Poor adaptability to complex environments. In practical applications, the accuracy of emotion recognition technology in complex environments (such as insufficient lighting, noise interference, facial occlusion) will be affected, thus reducing its practical utility in real-world scenarios (Paikrao et al., 2023).

9) Cross-cultural application barriers. Emotional expression patterns vary across diverse cultural backgrounds, and current emotion recognition systems often struggle to accurately understand and adapt to these differences when applied across-cultures, limiting their global deployment (Choi et al., 2022).

10) Algorithms have biases. As algorithms and models are trained on existing data that may harbor various biases and inequalities, the application of emotional artificial intelligence may exacerbate these issues and result in unfair treatment of certain groups. For example, biases may exist in sentiment analysis across different gender, race, and age groups (Alhelaly et al., 2025).

11) Autonomy is affected. If individuals overly rely on sentiment analysis techniques to understand and process their emotions, they may gradually lose control over their own emotions and become more reliant on external technologies and algorithms. This will undermine individual mental health and well-being, resulting in homogenized and stereotyped emotional expression and communication (Jordan, 2021).

6.2. Prospect of emotional privacy

The future research directions of emotional privacy based on artificial intelligence are diverse and of great significance (Bingyu Li et al., 2025). The following is a specific introduction.

1) Focusing on interdisciplinary integration and tackling key technologies in affective computing in education. The application of affective computing in the field of education combines the attributes of both educational and engineering disciplines. From the international development trend of affective computing education applications, its engineering attributes are becoming increasingly prominent (Dai et al., 2025). The current research hotspots in the application of affective computing technology in education, such as teaching evaluation based on affective computing and the application of affective agents, all require theories and methods from comprehensive interdisciplinary fields such as education, psychology, computer science, neuroscience, data science, cognitive science, behavioral science, and neuroscience (García-García et al., 2025).

2) Developing standards and specifications for the review of affective computing technology, achieving cross- machine emotional rule shaping. As affective computing is rapidly applied in education, it has a wider range of applications in online education, dual teacher teaching, special children's education, learning engagement assessment, and other scenarios. It is imperative to further establish and improve an open and transparent regulatory system for affective computing, and build a good regulatory environment for the innovative development of affective computing (Alhelaly et al., 2025).

3) Controlling the penetration of privacy boundaries and build a multi-dimensional collaborative governance under digital panoramic openness. In recent years, human society has gradually fallen into a "panoptic prison", giving rise to various "modern trust risks". Under digital panopticism, people hold a negative view of privacy, that is, we should actively abandon privacy and adopt a laissez-faire attitude toward privacy breaches (Epstein & Quinn, 2025). By utilizing digital technology to govern the application of affective computing in education, we can construct a long-term governance mechanism with platform connectivity, and scenario coordination through technology incremental empowerment and reconstruction innovation (Shao et al., 2024).

4) From the perspective of balancing tripartite interests, digital technology should be fully utilized to perceive social trends, assist scientific decision-making, and facilitate communication channels between government, enterprises, and schools. Enterprises are required to disclose the algorithm instructions related to affective computing without damaging their industry advantages and trade secrets, while also accepting public supervision, content review, and usability feedback. Based on the classification and grading mechanism of affective computing applications, the responsible parties will be further clarified, and the rights and responsibilities of both users and algorithm designers will be delineated to minimize ethical issues in affective computing (Satamraju & Balakrishnan, 2022).

5) Cutting-edge technology supports precise emotional modeling. Emotion and learning are inseparable components of human cognitive processes. Deeply interpreting the linkage effect and development laws of emotion and learning is of great significance for promoting individual learning and emotional development. Based on this, we consider the dynamics, individuality, and the relationship between emotions and behavior on existing emotion models to develop a more effective model to accurately describe the expression of learning emotions, and more precisely identify and analyze the emotional state and expression methods of individuals durning the learning process (Hidayat et al., 2024).

6) Develop lightweight non-invasive signal acquisition equipment. High-quality emotional data is the foundation of affective computing, however, high-quality emotional data is based on non-interference, real-time performance, and accuracy (Lai et al., 2024). Therefore, to support high-precision and multimodal emotional data collection, it is urgent to optimize sensor design from the aspects of device requirements analysis, hardware design, sensing technology, signal processing, and data analysis, and develop low-cost, lightweight, and high-precision emotional signal collection devices (Tsouvalas et al., 2022).

7) Situational orientation helps establish high-quality datasets. The current lack of high-quality datasets greatly limits the effective application of affective computing in education. It is urgent to

establish a high-quality large-scale emotional dataset for educational contexts to provide sufficient data support for the training and testing of affective computing algorithms (Eltanbouly et al., 2024). On the one hand, it is possible to artificially collect more data from real teaching contexts and fuse them to form a more comprehensive emotional dataset that spans multiple contexts and cultures, enhancing the model's understanding and adaptability to situational diversity and cultural differences (Hong, 2025).

8) Continuously iteration and optimization deep learning algorithms. Based on feature engineering and high-quality large-scale datasets, deep learning algorithm models are iteratively optimized through model tuning, parameter optimization, transfer learning, pre training, and other methods (Saber & Ahmed, 2024). On this basis, on the one hand, we adopt pre-training, data augmentation, and optimizer optimization methods to enhance the model's adaptability and generalization to different contexts, thereby improving its performance in affective computing tasks (Ntizikira et al., 2025).

9) Multi-stage and multimodal promotion of fusion technology innovation. The innovation of multimodal fusion technology needs to be based on specific tasks and data characteristics and cannot be generalized (Halkiopoulos et al., 2025). In practical applications, it is necessary to compare and select appropriate fusion methods based on the characteristics of tasks and datasets to attain optimal emotional state recognition performance. Researchers need to realize that although different modalities can achieve complementary information in emotion recognition, they also contain redundant components. In this regard, how to maintain the stability and effectiveness of models is also worth of attention and in-depth research (Ye et al., 2024).

10) Innovative affective computing scenario-based practices, such as integrating affective computing to improve immersive learning experiences, intelligent virtual learning companions. (Martin & Zimmermann, 2024). In terms of immersive learning experience, wearable device automatically perceives learners' emotions and obtains adaptive guidance and emotional arousal from intelligent learning systems. Regarding intelligent virtual learning companions, researchers have developed an intelligent mentor system integrated with intelligent emotional services (Neethirajan, 2024).

11) Unify privacy affective computing and artificial intelligence integrate technologies, such as cloud computing, federated learning, and differential privacy to achieve value mining of sentiment data while protecting user privacy (Jaiswal & Mower Provost, 2020). Using multimodal data for emotion recognition, combing homomorphic encryption to ensure that the data is processed under encrypted state, and implementing cross-institutional model training through federated learning, ensuring technical compliance and respecting user autonomy through ethical frameworks (Mendizabal-Ruiz & Paredes, 2024), as shown in Fig. 18.

7. Conclusion

Artificial intelligence empowers privacy emotional research, using advanced algorithms to analyze speech, facial expressions, and other data to identify emotional states and assist in personalized services in fields such as healthcare and education. However, emotional data contains sensitive information, and improper collection and processing can pose risks of privacy leakage. The core of privacy protection lies in the transparency and clarity of privacy boundaries.

Artificial intelligence technology should pay more attention to social reality and embed rational value thinking on human rights (such as privacy rights, personal information protection) and social development (Arman & Yang, 2024). When humans are unable to completely survive without technology, abandoning technological use means to some extent abandoning the right to life. Therefore, simply emphasizing the improvement of privacy literacy is insufficient to solve the problem (Yang et al., 2024).

In the future, it will be imperative to develop more secure privacy computing technologies, such as federated learning, differential privacy, to strike a balance between the value mining of emotional data and privacy protection (Patel & Hung, 2025).

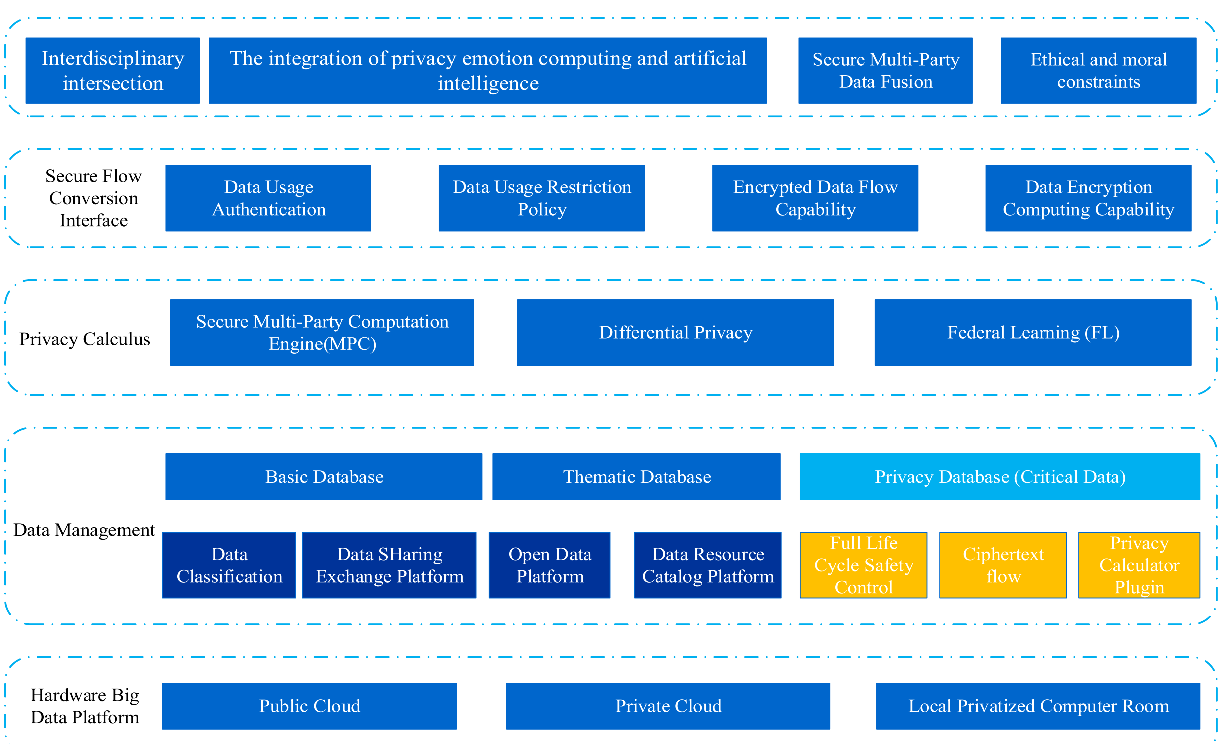

Fig. 18. Unified artificial intelligence and privacy emotion computing framework.

CRediT authorship contribution statement

Panjun Sun: Writing – review & editing, Writing – original draft, Conceptualization. **Hao Yu:** Writing – review & editing. **Wenjie Su:** Writing – review & editing. **Yule Wu:** Writing – review & editing.

Declaration of competing interest

I would like to declare that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part.

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